

Hao Vo

 [Website](#) |  [voanhhao3830 \(at\) gmail \(dot\) com](mailto:voanhhao3830@gmail.com) |  [Google Scholar](#)

RESEARCH INTERESTS

My research focuses on **4D scene understanding** - injecting geometric and temporal awareness into models from multi-view images and videos. My goal is to enabling robots to not only “see” but also “sense” the physical world. Longer term, I aim to help systems act reliably, advancing from task-specific solutions toward general-purpose, collaborative autonomy.

EDUCATION

University of Arkansas

Master, Computer Science - advised by Dr. Ngan Le

Arkansas, USA

May 2025 – May 2027

University of Information Technology - VNUHCM

B.S., Computer Science

HCM, Vietnam

Aug. 2021 – Dec. 2024

SELECTED PUBLICATIONS

SemLT3D: Semantic-Guided Expert Distillation for Camera-only Long-Tailed 3D Object Detection.

Hao Vo, Khoa Vo, Thinh Phan, Ngo Xuan Cuong, Gianfranco Doretto, Hien Nguyen, Anh Nguyen, Ngan Le
CVPR 2026

DRIVESPATIAL: A Benchmark for Spatiotemporal Intelligence in VLMs for Autonomous Driving.

Hao Vo, Khoa Vo, Phu Loc Nguyen, Sieu Tran, Duc Minh Nguyen, Ngo Xuan Cuong, Gladys Gawugah, Sreevenkata Anjani Tishita Godavarthi, Chase Rainwater, Nghi D. Q. Bui, Anh Nguyen, Duy Minh Ho Nguyen, Ngan Le
Under Review, 2026.

OmniSpace: Efficient Geometry Awareness for Autonomous Vehicles MLLMs.

Hao Vo, Phu Loc Nguyen, Khoa Vo, Sieu Tran, Duc Minh Nguyen, Ngo Xuan Cuong, Nghi D. Q. Bui, Anh Nguyen, Duy Minh Ho Nguyen, Ngan Le
Under Review, 2026.

Robust Motorcycle Helmet Detection in Real-World Scenarios: Using Co-DETR and Minority Class Enhancement.

Hao Vo, Sieu Tran, Duc Minh Nguyen, Thua Nguyen, Tien Do, Duy-Dinh Le
CVPRW 2024

MVDGC: Joint 3D and 2D Multi-view Pedestrian Detection via Dual Geometric Constraints.

Thinh Phan, **Hao Vo**, Khoa Vo, Thanh Ngo, Cuong Pham, Ngan Le
TMLR 2026

Dual-State Slot Attention: Decoupling Appearance and Identity for Video Object-Centric Learning.

Sieu Tran, Duc Nguyen, **Hao Vo**, Khoa Vo, Ngan Le
ACMMM 2026

EXPERIENCE

Graduate Research Assistant

University of Arkansas

May. 2025 - May. 2027

AR, USA

Research on multi-view understanding and spatial intelligence for autonomous systems, spanning self-driving cars and robotics, with papers published and under submission at venues such as CVPR, NeurIPS, and TMLR.

- **Self-driving cars.** Lifted the generalization of vision-language models on long-tail driving distributions and strengthened their spatial reasoning, producing three first-author papers.
- **Robot manipulation.** Studied physical understanding through extrinsic dexterity, with an ongoing paper in progress.

AI Engineer

viAct

Sep. 2023 - Apr. 2025

Hong Kong

Built computer vision for construction safety and led AI deployment across Singapore and KSA, covering 20+ projects and 300+ cameras.

- **3D computer vision.** Designed a 3D-aware method to detect people inside suspended-load danger zones and optimized it to run 10 cameras at real-time FPS on Jetson NX, while leading a team of 8 on algorithm design and research direction.
- **MLOps.** Designed an end-to-end pipeline for data collection, labeling, training, and deployment, cutting data-processing time by 30%.

Undergraduate Research Assistant

Sep. 2021 - May. 2023

University of Information Technology - VNUHCM

HCM, Vietnam

Research on 2D perception and multimodal video systems for real-world applications.

- **2D perception.** Designed detection algorithms robust to extreme scenarios, resulting in a CVPRW 2024 paper.
- **Video retrieval.** Built a system that retrieves a ~10-second event from 1.3 TB of video in 0.61s using speech, text, or image queries, which won the largest AI competition in Vietnam.
- **Face detection.** Helped build a face-recognition system deployed for classroom attendance checking at UIT.

AWARDS

First Prize, AI City Challenge (Track 5), **NVIDIA**. Highest performance at the CVPR Workshop. (2024)

First Prize, Ho Chi Minh City AI Challenge, Vietnam. Best-performing event retrieval system. (2022)

Second Prize, Vietnam Ministerial-level Scientific Research. Helmet-violation detection for motorbike riders. (2025)

Second Prize, FPT Vietnam AI Challenge. Runner-up Lost-and-Found system. (2023)

Third Prize, BKAI-Naver Challenge. Our method recognized as top-3 for scene text recognition. (2021)

Consolation Prize, UIT Game Hackathon. Computer vision system for environmental protection. (2022)

Finalist, Zalo AI Challenge. Top-5 in adversarial personalized image generation. (2024)

TECHNICAL SKILLS

Languages: Python, JavaScript, C/C++, Go.

Frameworks: Torch, React, Node.js, Next.js, FastAPI.

Developer Tools: Git, Docker, Airflow, Ansible, Vault, Signoiz, MLflow, Celery, Redis, RabbitMQ, Nginx, Slurm, AWS, GitCI, MongoDB, SQL, DVC.